Mona Pouresmaeil

MSc Computer Engineering – AI and Data Analytics

Turin, Piedmont, Italy

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Professional Summary

MSc Computer Engineering student (expected graduation July 2026) specializing in **Machine Learning**, **Deep Learning**, **NLP**, and **LLMs**. Currently completing my **AI Engineer Internship** at **Brain Technologies**, where I am finalizing my Master's thesis on advanced **Retrieval-Augmented Generation (RAG)** pipelines and **NLP** applications for document validation.

Co-author of the **SemEval 2026 Task 1** system that ranked **2nd internationally** (accepted paper). Hands-on experience with **LangChain**, **LangGraph**, **Llama 3**, and **agentic AI**, backed by strong **Python** and **Django** skills.

Actively seeking full-time **AI/ML Engineer** or **NLP Engineer** roles starting summer 2026.

Education

Politecnico di Torino *Turin, Italy* 2023 – July 2026

- **MSc in Computer Engineering** – Artificial Intelligence & Data Analytics Track

Shariaty Technical and Vocational College *Tehran, Iran* 2018 – 2022

- **BSc in Computer Technology Engineering (Software)** — 2020 – 2022
- **Associate Degree in Computer Software** — 2018 – 2020

Experience

AI Engineer Intern *Brain Technologies — Turin, Italy (On-site)* January 2026 – Present

- Developing Master's thesis: building an AI framework for automatic search, validation, and verification of documentary content using **RAG**, **NLP techniques**, and **LLM-based pipelines**
- Designed and implemented RAG architectures with LangGraph workflows, ChromaDB, confidence scoring for hallucination mitigation, and compliance features
- Gaining practical experience in production-oriented LLM applications and information retrieval systems

Backend Developer Intern *Rastava Academy — Remote* March 2020 – August 2020

- Developed scalable backend services using **Python and Django**
- Collaborated in Agile teams with Git/GitHub for version control and code reviews
- Designed and optimized database schemas for data integrity and efficient retrieval

Software Engineering Intern *Kasra Company — Tehran, Iran (On-site)* September 2019 – February 202

- Customized enterprise software solutions for 500+ clients according to specific business needs and workflows
- Optimized SQL queries to improve database performance and reduce latency
- Participated actively in Agile development cycles

Key Projects

Agentic Job Tracker 2026

(GitHub)

- Built an AI-powered tool that processes Italian job postings with **auto-translation** (IT→EN), **intelligent field extraction** (company, location, RAL, tech stack), Italian tax estimator, and an **agentic chat interface**
- **Tech:** Python 3.13, Streamlit, Llama 3.3-70B-Instruct, Hugging Face, Agentic AI

SemEval 2026 Task 1 – MWAHAHA (Humor Generation)

(GitHub)

- Co-developed a humor generation system that achieved **2nd place** internationally
- Co-authored the accepted **system description paper** (pending publication)
- **Tech:** Python, NLP, Large Language Models

Fake News Detection System

(GitHub)

- Created a complete NLP pipeline including text preprocessing, **TF-IDF**, and classification models (**SVM**, **Logistic Regression**, **LSTM**)
- **Tech:** Python, scikit-learn, NLTK, LSTM

Deep Learning Handwritten Digit Classifier

(GitHub)

- Implemented an advanced **CNN** in **PyTorch** with **batch normalization**, **dropout**, and **data augmentation**; achieved **99%+ accuracy** on MNIST dataset
- **Tech:** PyTorch, Computer Vision

Skills

- **AI & Machine Learning:** Machine Learning, Deep Learning, NLP, LLMs, Retrieval-Augmented Generation (RAG), Agentic AI, Feature Engineering, Model Evaluation, Hallucination Mitigation
- **Frameworks & Libraries:** Python, PyTorch, scikit-learn, LangChain, LangGraph, Hugging Face, TensorFlow, XGBoost, Pandas, NumPy, spaCy, NLTK, Ragas
- **Software Engineering & Tools:** Django, SQL, Query Optimization, React, REST APIs, Git/GitHub, Docker, AWS, Streamlit, Flask, Agile, Jupyter

Publications & Achievements

- **SemEval 2026 Task 1 (MWAHAHA) – System Description Paper**
  Ranked **2nd** internationally • Accepted (pending publication)
  Co-author with professors and teammates

Certifications

- RAG for Professionals with LangGraph, Python and OpenAI – Udemy (Dec 2025)
- Introduction to TensorFlow for Artificial Intelligence, Machine Learning, and Deep Learning – DeepLearning.AI (Nov 2025)
- GenAI Intensive Program – Kaggle

Languages

- **English** — Fluent
- **Italian** — Limited working proficiency (A2/B1)